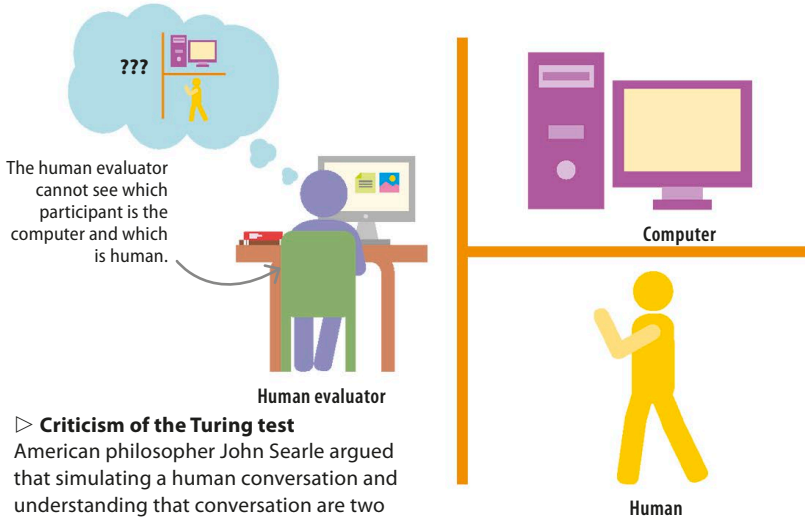


The Turing test

In 1950, English scientist Alan Turing invented a test of artificial intelligence. The test involves an evaluator who is able to ask a computer and a human questions by text, and they are able to reply. The evaluator doesn't know which of the responses are from the human and which are from the computer. If the evaluator cannot tell which is the machine more than 50 per cent of the time, then the machine has passed the test and is said to be intelligent.



► Criticism of the Turing test

American philosopher John Searle argued that simulating a human conversation and understanding that conversation are two different things. Humans also have a lot of unintelligent behaviour, such as making spelling mistakes that might influence a human evaluator during the Turing test.

IN DEPTH

Issues

Allowing AI to make its own decisions might seem safe in a controlled environment, but if intelligent machines were to be used in a wider way, they could be involved in real-world life-and-death situations.

- **Ethical dilemmas:** A self-driving car hits a patch of ice and skids out of control. The car can either swerve into a wall and kill the passenger inside, or veer into a crowd, saving the passenger but potentially killing more people. Before, this decision rested with the human driver, but now it rests with the programmers of the self-driving car. The programmers also have another ethical dilemma in that their customers are unlikely to want to buy a car that will prioritize the safety of others over them – which may make the programmers lean towards saving the passenger at other people's expense.

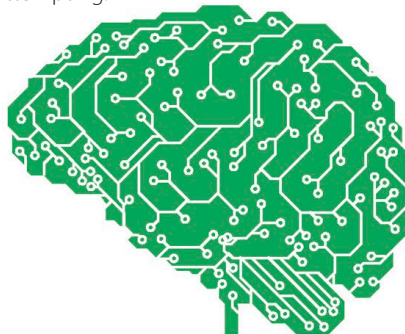
- **Bad data:** If an AI is fed the wrong data, or if there are errors in the data or the way the AI is designed, it can make bad decisions. It can be difficult to identify and locate these problems.

General intelligence

Another goal of AI is to create a general intelligence capable of learning different types of information instead of one specific task. Logic AI, also known as classical AI, represents knowledge using symbols. It takes a top-down approach to intelligence by defining formulas that the computer can use to solve varied problems. A general intelligence AI will be able to figure out when something it already knows is relevant and useful to complete whatever task it is currently attempting.

► Artificial neural networks (ANN)

ANN algorithms are a kind of general intelligence, and are loosely inspired by real brains. Layers of interconnected digital "neurons" manipulate numerical data using statistical equations.



"The greatest danger of artificial intelligence is that people conclude too early that they understand it."

Eliezer Yudkowsky (b. 1979), AI researcher and writer

IN DEPTH

Why now?

An increase in computing power has allowed increasingly sophisticated AI to be developed. Neural networks have been around since the 1970s, but until recently they took too much time to train. AI studies will affect the real world more and more as people continue to develop computer chips with ever-greater processing power.